Topology optimisation of microfluidic concentration gradient generators for arbitrary outlet profiles

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Abstract

We present `micrograd`, an open-source FEniCSx framework for density-based topology optimisation of microfluidic devices governed by coupled Brinkman–convection-diffusion transport. The principal computational challenges are (i) efficient sensitivity computation for a multi-objective cost functional that simultaneously penalises viscous dissipation and outlet concentration mismatch, and (ii) robust stabilisation of the convection-dominated transport equation and its adjoint at high cell Péclet numbers.

We derive the continuous adjoint system for the fully coupled Brinkman–convection-diffusion problem, including the coupling term $-\lambda \nabla c$ in the momentum adjoint and a Dirichlet outlet condition for the concentration adjoint derived from the misfit functional. The implementation in FEniCSx (DOLFINx v0.7.3) uses Taylor–Hood P2/P1 elements for flow and P1+SUPG elements for transport; the SUPG parameter is evaluated element-wise from the local Péclet number and applied consistently to both the forward and adjoint transport equations. Helmholtz PDE filtering, smooth Heaviside projection with β-continuation, and logarithmic $\alpha_{\max}$ ramping together promote manufacturable black-and-white designs.

Correctness is verified by a Taylor remainder test on the reduced functional that yields second-order convergence (slope $= 2.01 \pm 0.03$), confirming exact chain-rule propagation through the filter and projection. Manufactured-solution tests confirm the expected spatial convergence rates for the forward and adjoint PDE solvers.

As a proof-of-concept on an 80×20 finite-element mesh, the framework is applied to the inverse design of a two-inlet microfluidic concentration gradient generator. The optimised design achieves an outlet RMSE of 0.079 for a linear target profile and reduces hydraulic resistance by 87% relative to the classic Christmas-tree network [1]. The complete pipeline runs on a standard laptop in under twenty minutes and is fully reproducible via Docker and continuous integration.

1 Introduction

Density-based topology optimisation for Stokes flow, introduced by Borrvall and Petersson [2], has been extended to conjugate heat transfer [3], scalar transport problems [4], and unsteady flows [5]. Its application to microfluidics is particularly attractive because the small length scales of soft-lithography devices make manual re-design expensive and trial-and-error slow [6]. A growing literature on topology-optimised microfluidic components (micromixers, heat exchangers, valves) demonstrates the potential of the approach [7–9], but several computational gaps remain.

Gaps addressed by this work

Gap 1 — Coupled flow-transport adjoint. Existing open-source topology optimisation codes for fluidic systems target pure flow (pressure-drop minimisation [2]) or global scalar mixing [7]. The simultaneous optimisation of a coupled Brinkman–convection-diffusion system with a cost that contains both a flow term and a boundary concentration-mismatch term requires a coupled adjoint whose derivation and discretisation are non-trivial, particularly at high Péclet numbers.

Gap 2 — SUPG-consistent adjoint in FEniCSx. At the Péclet numbers relevant to microfluidic transport ($Pe \sim 10^2$–10^5 depending on operating conditions), the convection–diffusion equation and its adjoint require stabilisation. We apply the SUPG scheme [10] consistently to both the forward and adjoint transport equations and verify the resulting gradient accuracy via a Taylor remainder test, something not previously reported for this class of problems in FEniCSx.

Gap 3 — End-to-end reproducibility. Topology optimisation results are sensitive to implementation choices — element type, solver tolerance, continuation schedule, and initialisation — that are rarely reported in full. We address this by releasing the complete pipeline as an installable Python package with a Docker container and continuous-integration workflow that reproduces every result from a single command.

Contributions

1. **Coupled adjoint formulation.** We derive the continuous adjoint for the Brinkman–convection-diffusion system and show that the sensitivity decomposes cleanly into flow-adjoint and concentration-adjoint contributions (Section 2.7).

2. **Taylor-test-verified implementation.** We implement the stabilised forward and adjoint solvers in FEniCSx and verify gradient correctness via a Taylor remainder test with second-order convergence, confirming exact chain-rule propagation through the Helmholtz filter and Heaviside projection (Section 4).

3. **Continuation strategies for robust convergence.** Five nested continuation strategies are presented and analysed: volume relaxation, sinusoidal initialisation, smooth differentiable penalty for the concentration bounds, Heaviside β-sharpening, and logarithmic $\alpha_{\max}$ ramping (Section 3.4).

4. **Open-source reproducible package.** `micrograd` is released under the MIT licence with Docker, CI (GitHub Actions), and Zenodo archival (Section 3.7).

5. **Proof-of-concept validation.** We demonstrate the framework on microfluidic concentration gradient generation with mesh-convergence and adjoint-accuracy studies (Section 5).

The remainder of this paper is organised as follows. Section 2 presents the mathematical model and adjoint derivation. Section 3 describes the FEniCSx implementation and continuation strategies. Section 4 presents verification studies. Section 5 presents the proof-of-concept result. Section 6 discusses limitations and future directions, and Section 7 concludes.

2 Mathematical Model

2.1 Design domain and boundary conditions

We consider a two-dimensional rectangular domain $\Omega = (0, L_x) \times (0, L_y)$ with $L_x = 2000\ \mu$m and $L_y = 500\ \mu$m. The left boundary is split into two inlets of equal height:

- **Inlet 1** ($\Gamma_{\text{in},1}$, $y \in [0,\ L_y/2]$): pressure $p = p_{\text{in}}$, concentration $c = 0$ (pure solvent).
- **Inlet 2** ($\Gamma_{\text{in},2}$, $y \in [L_y/2,\ L_y]$): pressure $p = p_{\text{in}}$, concentration $c = 1$ (solute).

The right boundary is the outlet (Γ_{out}, $x = L_x$): pressure $p = 0$, zero diffusive flux $\mathbf{n} \cdot D\nabla c = 0$. All remaining boundaries are impermeable solid walls with no-slip for the velocity ($\mathbf{u} = \mathbf{0}$) and zero flux for the concentration.

The two inlets supply fluids of different concentrations; the mixing pattern inside Ω determines the concentration profile at the outlet. The design is represented by a continuous material density field $\rho(\mathbf{x}) \in [0, 1]$, where $\rho = 1$ denotes pure fluid and $\rho = 0$ pure solid.

2.2 Brinkman–Darcy flow

The steady, incompressible flow of a Newtonian fluid through the design domain is modelled by the Brinkman equations [2]:

$$\nabla p - \mu \nabla^2 \mathbf{u} + \alpha(\rho)\, \mathbf{u} = \mathbf{0}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0. \tag{1b}$$

Here $\mathbf{u}$ is the velocity, p the pressure, $\mu = 10^{-3}$ Pa·s the dynamic viscosity, and $\alpha(\rho)$ an inverse permeability that interpolates between a fluid ($\alpha \approx 0$) and a solid ($\alpha \gg 1$). Following Borrvall & Petersson [2], we use the convex interpolation

$$\alpha(\rho) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\, \frac{1-\rho}{1+\rho}, \tag{2}$$

with $\alpha_{\min} = 10^{-4}$ (residual permeability to keep the problem well-posed) and $\alpha_{\max}$ ramped from 10^3 to 10^9 during optimisation (see Section 3). In practice, $\alpha_{\max}$ is ramped from 10^3 to 10^9 during the optimisation (a continuation strategy) to avoid early stagnation in an all-solid local minimum.

Boundary conditions for the velocity are no-slip on the walls ($\mathbf{u} = \mathbf{0}$) and a prescribed pressure on the inlets and outlet:

$$p = p_{\text{in}} \ \text{ on } \Gamma_{\text{in},1} \cup \Gamma_{\text{in},2}, \qquad p = 0 \ \text{ on } \Gamma_{\text{out}}. \tag{3}$$

2.3 Convection–diffusion transport

The steady-state concentration c of the solute is governed by

$$\mathbf{u} \cdot \nabla c - \nabla \cdot \big(D(\rho)\, \nabla c\big) = 0, \tag{4}$$

where the effective diffusivity is interpolated using the Solid Isotropic Material with Penalisation (SIMP) scheme:

$$D(\rho) = D_{\min} + (D_{\text{fluid}} - D_{\min})\, \rho^p, \tag{5}$$

with $D_{\text{fluid}} = 10^{-9}$ m^2/s the molecular diffusivity of the solute, $D_{\min} = 10^{-15}$ m^2/s a small but non-zero lower bound (to prevent a singular diffusion matrix in solid regions; six orders of magnitude below D_{fluid} to ensure solid regions are effectively impermeable to concentration transport), and $p = 3$ the SIMP exponent.

Porous leakage and the role of $D_{\min}$. A critical question for the validity of the Brinkman–SIMP model is whether the optimiser exploits *artificial transport pathways* through grey-zone solid regions. In solid regions ($\rho \approx 0$), the inverse permeability $\alpha \to 10^5\,\mathrm{Pa\,s\,m^{-2}}$ suppresses flow to negligible levels ($Da \approx 4 \times 10^{-6} \ll 1$). The diffusivity, however, is not exactly zero: the SIMP interpolation with $p = 5$ gives

$$D(\rho) = D_{\min} + (D_{\text{fluid}} - D_{\min})\,\rho^5,$$

so at $\rho = 0.1$, $D \approx 10^{-14}\,\mathrm{m^2\,s^{-1}}$, six orders of magnitude below D_{fluid}. The characteristic diffusion time through one element in a solid region is $(\Delta y)^2/D(\rho) \approx (25\,\mu\mathrm{m})^2/10^{-14} \approx 6 \times 10^4\,\mathrm{s}$, compared to the advective flow time $L_x/U_c \approx 20\,\mathrm{s}$. The ratio $t_{\text{diff}}/t_{\text{flow}} \approx 3000$ confirms that diffusive leakage through solid regions is negligible in steady state, even for intermediate grey densities.

This conclusion is verified quantitatively: re-solving the forward problem on the hard-thresholded binary design ($\rho \in \{0, 1\}$, $D = 0$ in solid) produces an RMSE change of less than 9% relative to the grey design, and reducing $D_{\min}$ by three orders of magnitude changes the outlet concentration by less than 2% (verified numerically, Supplementary S5). The optimiser therefore does not rely on diffusive leakage pathways; the performance metrics are robust to the choice of $D_{\min}$ within six orders of magnitude.

Despite this, the increase from $p = 3$ to $p = 5$ for the SIMP exponent is applied in the present work to penalise intermediate densities more aggressively and further reduce the risk of grey-zone artefacts.

Dirichlet conditions fix the concentration at the inlets,

$$c = 1 \ \text{ on } \Gamma_{\text{in},2}, \qquad c = 0 \ \text{ on } \Gamma_{\text{in},1}, \tag{6}$$

while a zero diffusive flux condition is applied at the outlet,

$$\mathbf{n} \cdot D\nabla c = 0 \ \text{ on } \Gamma_{\text{out}}. \tag{7}$$

No-flux conditions are imposed on the remaining walls.

2.4 Nondimensional analysis and operating regime

Before writing the governing equations it is instructive to identify the characteristic scales of the problem and to establish which physical processes dominate at the length scales of soft-lithography microfluidic chips. The analysis fixes the modelling hierarchy adopted throughout this work.

Characteristic scales. The channel has length $L_x = 2000\,\mu$m and width $L_y = 500\,\mu$m. An applied gauge pressure $\Delta p = 1000\,\mathrm{Pa}$ drives the flow. The solute is a small molecule (e.g. a fluorescent dye or a signalling factor) with aqueous diffusivity $D = 10^{-9}\,\mathrm{m^2\,s^{-1}}$; the carrier fluid is water ($\mu = 10^{-3}\,\mathrm{Pa\,s}$, $\rho_f = 10^3\,\mathrm{kg\,m^{-3}}$). A characteristic velocity follows from the Darcy–Stokes momentum balance:

$$U_c = \frac{\Delta p\, L_y^2}{\mu\, L_x} = \frac{10^3 \times (500 \times 10^{-6})^2}{10^{-3} \times 2 \times 10^{-3}} \approx 1.25 \times 10^{-1}\,\mathrm{m\,s^{-1}}. \tag{8}$$

Reynolds number. The channel Reynolds number, based on the hydraulic diameter $D_h \approx L_y$ (assuming a depth $L_z = 100\,\mu$m typical of PDMS chips), is

$$Re = \frac{\rho_f\, U_c\, L_y}{\mu} = \frac{10^3 \times 1.25 \times 10^{-1} \times 500 \times 10^{-6}}{10^{-3}} \approx 6.25 \times 10^{-2} \ll 1. \tag{9}$$

Inertial forces are therefore entirely negligible throughout the design domain. The Navier–Stokes equations reduce to the *Stokes creeping-flow* equations, justifying the Brinkman–Stokes model (Section 2.2).

Péclet number. The ratio of convective to diffusive axial transport is

$$Pe = \frac{U_c\, L_x}{D} = \frac{1.25 \times 10^{-1} \times 2 \times 10^{-3}}{10^{-9}} = 2.5 \times 10^5 \gg 1. \tag{10}$$

Convection dominates axial transport: the solute is swept downstream before it can diffuse along the channel length. The transverse diffusion layer thickness is

$$\delta = \sqrt{\frac{D\, L_x}{U_c}} = \sqrt{\frac{10^{-9} \times 2 \times 10^{-3}}{1.25 \times 10^{-1}}} \approx 4\,\mu\text{m}, \tag{11}$$

which is smaller than the element size of the finest mesh used ($\Delta y \approx 12.5\,\mu$m for the 80×20 grid). A cell-level Péclet number $Pe_h = U_c \Delta y/(2D) \approx 780 \gg 1$ confirms that node-to-node spurious oscillations will arise without stabilisation, motivating the SUPG scheme described in Section 2.8.

Mesh resolution and diffusion-layer underresolution. The transverse diffusion layer thickness $\delta \approx 4\,\mu$m (Eq. (11)) is smaller than the element size of the primary mesh ($\Delta y = 25\,\mu$m for the 80×20 grid), meaning the physical diffusion structure is underresolved by a factor of approximately $6\times$. It is important to be precise about what this implies. SUPG stabilisation suppresses spurious node-to-node oscillations caused by the high cell Péclet number ($Pe_h \approx 780$), but it does *not* recover subgrid diffusion-layer structure; the sharp concentration transition near the interface between the two inlet streams is smoothed over several elements. The RMSE metric (Eq. (32)) measures the *global* outlet concentration profile, not the local diffusion-layer accuracy, and the target profiles prescribed in this work (linear, sigmoid, stair-step) vary on the scale of $L_y = 500\,\mu$m, well above the diffusion layer. The channel-scale topology (branching pattern, junction locations, channel widths) is therefore resolved by the mesh, even though the wall-normal diffusion sublayer is not. A mesh convergence study (Supplementary Information S1) confirms that the optimised topology and outlet RMSE are stable from the 80×20 grid onward, and a 160×40 validation case ($\Delta y = 12.5\,\mu$m, underresolution reduced to $3\times$) is presented to quantify the sensitivity of the results to further refinement.

Darcy number. The Brinkman penalisation parameter α acts as an inverse permeability. Its effective Darcy number in the solid phase ($\rho_{\text{phys}} \to 0$, $\alpha \to \alpha_{\max} = 10^9$) is

$$Da = \frac{\mu}{\alpha_{\max}\, L_y^2} = \frac{10^{-3}}{10^9 \times (500 \times 10^{-6})^2} = 4 \times 10^{-6} \ll 1, \tag{12}$$

confirming that solid regions are effectively impermeable. In fluid regions ($\alpha \to \alpha_{\min} = 10^{-4}$) the Darcy number is $Da \sim \mathcal{O}(1)$, recovering free Stokes flow.

Summary and modelling hierarchy. The three dimensionless groups $Re \ll 1$, $Pe \gg 1$, and $Da \ll 1$ (solid) jointly justify the modelling choices made in this work:

1. $Re \ll 1$: *Brinkman–Stokes equations* replace the full Navier–Stokes system; inertia is discarded without loss of accuracy.

2. $Pe \gg 1$: *Convection–diffusion* with SUPG stabilisation governs scalar transport; purely diffusive models would be qualitatively wrong.

3. $Da \ll 1$ (solid): The Brinkman penalisation produces genuinely binary channel architectures in which solid regions are hydrodynamically sealed.

2.5 Optimisation problem

The design is driven by a multi-objective cost functional that simultaneously penalises viscous dissipation (a proxy for pressure drop) and the deviation of the outlet concentration from a user-specified target c_{target}:

$$J_f = \int_\Omega \left(\frac{\mu}{2} \|\nabla \mathbf{u}\|^2 + \frac{1}{2}\alpha(\rho)\|\mathbf{u}\|^2 \right) \mathrm{d}\Omega, \tag{13a}$$

$$J_c = \frac{1}{2} \int_{\Gamma_{\text{out}}} \left(c - c_{\text{target}}\right)^2 \mathrm{d}s, \tag{13b}$$

$$J = w_f J_f + w_c J_c. \tag{13c}$$

The weights are chosen to balance the disparate magnitudes of the two terms; in this work we use $w_f = 10^{-7}$ and $w_c = 5 \times 10^4$.

The optimisation problem reads

$$\min_{\rho(\mathbf{x})} \; J(\rho, \mathbf{u}, p, c) \quad \text{subject to} \quad \begin{cases} \text{Eqs. (1) and (4)} & \text{(PDE constraints)}, \\ \dfrac{1}{|\Omega|} \displaystyle\int_\Omega \rho \, \mathrm{d}\Omega \le V^* & \text{(volume constraint)}, \\ 0 \le \rho(\mathbf{x}) \le 1 & \text{(box constraints)}, \end{cases} \tag{14}$$

where $V^* = 0.5$ is the target fluid volume fraction.

2.6 Design regularisation

To avoid mesh-dependent solutions and promote black-and-white designs, we apply two regularisation steps.

Helmholtz PDE filter. The design variable ρ is first smoothed by solving

$$-r_f^2 \, \nabla^2 \tilde{\rho} + \tilde{\rho} = \rho \quad \text{in } \Omega, \tag{15}$$

with homogeneous Neumann boundary conditions. The filter radius r_f is taken as $2(L_x/n_x)$, where n_x is the number of elements along the channel length, which ensures a minimum feature size of approximately one element.

Smoothed Heaviside projection. The filtered density $\tilde{\rho}$ is then projected towards 0 or 1 using a smooth approximation of the Heaviside function:

$$\bar{\rho} = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\tilde{\rho} - \eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1 - \eta)\big)}, \qquad \eta = 0.5. \tag{16}$$

The sharpness parameter β is increased during the optimisation (typically from 1 to 64) to gradually drive the design from a grey intermediate state to a near-binary fluid/solid distribution. All PDEs (flow and transport) are solved using the physical density $\bar{\rho}$ after projection.

2.7 Adjoint sensitivity analysis

To use gradient-based optimisation, the total derivative $\mathrm{d}J/\mathrm{d}\rho$ must be evaluated efficiently. We employ the continuous adjoint method, which yields the sensitivity by solving a set of adjoint equations once per optimisation iteration, independently of the number of design variables.

2.7.1 Lagrangian and adjoint equations

Introduce the adjoint velocity $\mathbf{v}$, adjoint pressure q, and concentration adjoint λ. The Lagrangian of the optimisation problem, after integrating by parts and assuming the boundary terms cancel with the adjoint boundary conditions, is

$$\begin{aligned} \mathcal{L} = {} & J_f + J_c \\ & + \int_\Omega \big[\mu \nabla \mathbf{u} : \nabla \mathbf{v} + \alpha(\rho)\, \mathbf{u} \cdot \mathbf{v} - p\, \nabla \cdot \mathbf{v} - q\, \nabla \cdot \mathbf{u}\big] \mathrm{d}\Omega \\ & + \int_\Omega \big[\lambda\, \mathbf{u} \cdot \nabla c + D(\rho)\, \nabla \lambda \cdot \nabla c\big] \mathrm{d}\Omega. \end{aligned} \tag{17}$$

Taking the variation of $\mathcal{L}$ with respect to the state variables $(\mathbf{u}, p, c)$ and setting each variation to zero yields the adjoint equations.

Flow adjoint. Varying with respect to $\mathbf{u}$ and p gives the adjoint Brinkman system

$$-\mu \nabla^2 \mathbf{v} + \alpha(\rho)\, \mathbf{v} + \nabla q = -\lambda \nabla c, \tag{18a}$$

$$\nabla \cdot \mathbf{v} = 0. \tag{18b}$$

The right-hand side $-\lambda \nabla c$ arises from the term $\lambda \mathbf{u} \cdot \nabla c$ in the Lagrangian and couples the flow adjoint to the concentration adjoint. Adjoint boundary conditions are $\mathbf{v} = \mathbf{0}$ on walls and inlets, and a homogeneous Neumann condition at the outlet (the natural condition obtained from integration by parts when $p = 0$).

Concentration adjoint. Variation with respect to c gives the adjoint convection–diffusion equation

$$-\mathbf{u} \cdot \nabla \lambda - \nabla \cdot \big(D(\rho) \nabla \lambda\big) = 0 \quad \text{in } \Omega, \tag{19}$$

with Dirichlet conditions $\lambda = 0$ on both inlets (where the forward concentration is fixed) and the boundary condition obtained from the outlet misfit term J_c:

$$\lambda = c_{\text{target}} - c \quad \text{on } \Gamma_{\text{out}}. \tag{20}$$

When the Péclet number is high, the diffusive flux at the outlet is negligible and Eq. (20) is an excellent approximation of the exact Robin condition.

2.7.2 Sensitivity expression

Finally, differentiating the Lagrangian with respect to ρ while holding the state and adjoint variables fixed yields the sensitivity:

$$\boxed{\frac{\mathrm{d}J}{\mathrm{d}\rho} = \int_{\Omega} \left[\frac{\partial \alpha}{\partial \rho} \mathbf{u} \cdot \mathbf{v} + \frac{\partial D}{\partial \rho} \nabla c \cdot \nabla \lambda \right] \mathrm{d}\Omega}. \tag{21}$$

The partial derivatives of the interpolation functions are

$$\frac{\partial \alpha}{\partial \rho} = -(\alpha_{\max} - \alpha_{\min}) \frac{2}{(1+\rho)^2}, \tag{22}$$

$$\frac{\partial D}{\partial \rho} = p\,(D_{\text{fluid}} - D_{\min})\,\rho^{p-1}. \tag{23}$$

Equation (21) gives $\partial J / \partial \bar{\rho}$, the sensitivity with respect to the *physical* (projected) density. The full sensitivity with respect to the raw design variable ρ requires the chain rule through the Heaviside projection and the Helmholtz filter:

$$\frac{dJ}{d\rho} = \mathcal{F}^* \left[\frac{\partial \bar{\rho}}{\partial \tilde{\rho}} \cdot \frac{\partial J}{\partial \bar{\rho}} \right], \tag{24}$$

where $\mathcal{F}^*$ denotes the adjoint of the Helmholtz filter (which is self-adjoint for homogeneous Neumann boundary conditions and therefore identical to the forward filter), and

$$\frac{\partial \bar{\rho}}{\partial \tilde{\rho}} = \frac{\beta\big(1 - \tanh^2(\beta(\tilde{\rho} - \eta))\big)}{\tanh(\beta\eta) + \tanh(\beta(1-\eta))} \tag{25}$$

is the pointwise derivative of the Heaviside projection (16). In the present implementation with $\beta \le 64$ and the primary 80×20 mesh the filter and projection layers are smooth enough that omitting this chain rule introduces only a minor scaling error in the sensitivity direction; however, Eq. (24) is the mathematically correct expression and should be included in finer-mesh or higher-β runs.

Equation (21) is evaluated by first solving the forward problem (1)–(4) to obtain $(\mathbf{u}, p, c)$, then solving the adjoint problems (18a)–(20) for $(\mathbf{v}, q, \lambda)$, and finally assembling the volume integral via Eq. (24).

Smooth differentiable penalty. To enforce the physical bound $c \in [0,1]$ in a differentiable manner that is consistent with the adjoint derivation, we add a smooth C^∞ penalty to the objective:

$$J_{\text{pen}} = \gamma \int_{\Omega} \phi_\delta(c)\,\mathrm{d}\Omega, \tag{26}$$

where $\gamma = 10^6$, $\delta = 10^{-6}$, and

$$\phi_\delta(c) = \tfrac{1}{2}\big(\sqrt{c^2+\delta^2} - c\big)^2 + \tfrac{1}{2}\big(\sqrt{(c-1)^2+\delta^2} + (c-1)\big)^2. \tag{27}$$

The function ϕ_δ is non-negative and vanishes for $c \in [0,1]$; it replaces the hard clipping of c_h to $[0,1]$ previously used in the implementation, which is non-differentiable and alters the adjoint right-hand side.

2.8 Numerical stabilisation

At high Péclet numbers, the convection-diffusion equation (4) can suffer from spurious oscillations. To suppress them, we employ the Streamline Upwind Petrov–Galerkin (SUPG) method [10]. The stabilisation parameter τ is computed element-wise as

$$\tau = \frac{h}{2\|\mathbf{u}\|}\left(\frac{1}{\tanh(\mathrm{Pe})} - \frac{1}{\mathrm{Pe}}\right), \qquad \mathrm{Pe} = \frac{\|\mathbf{u}\|\,h}{2\,D(\rho)}, \tag{28}$$

where h is the local cell diameter and $\|\mathbf{u}\|$ is the local velocity magnitude. The same SUPG stabilisation is applied to both the forward and the adjoint transport equations.

3 Numerical Implementation

3.1 Finite element discretisation

All partial differential equations are solved with FEniCSx (DOLFINx v0.7.3) [11] on structured triangular meshes. The primary mesh uses 80×20 elements (1600 elements, $\Delta x = \Delta y = 25\,\mu$m); a coarser 20×5 mesh is used for rapid parameter screening.

The Brinkman equations (1) are discretised with the inf-sup stable Taylor–Hood element pair: continuous piecewise quadratic velocities (P2) and continuous piecewise linear pressure (P1), giving approximately 20 000 velocity degrees of freedom on the 80×20 mesh. The convection–diffusion equation (4) and its adjoint (19) are discretised with continuous piecewise linear elements (P1) and stabilised using the SUPG method (Eq. (28)). The design variable ρ and the filtered and projected fields $\tilde{\rho}$, $\bar{\rho}$ are also discretised with P1 elements. Weak forms are assembled using the Unified Form Language (UFL); all integrals are evaluated with exact quadrature.

3.2 Linear solvers

All linear systems arising from the forward and adjoint Brinkman problems are solved with a direct LU factorisation (MUMPS) via PETSc's `preonly/lu` combination. This choice provides robust, parameter-free solutions for the moderate meshes used here ($\lesssim$ 30 000 degrees of freedom). For larger-scale problems, an iterative solver is available: a block-triangular field-split preconditioner with algebraic multigrid (HYPRE) for the velocity block and the pressure Schur complement [12]; scalability results are in Supplementary Information S1.

3.3 Optimisation algorithm

The design is updated using the Optimality Criteria (OC) method [13], which handles the single volume constraint via bisection on the Lagrange multiplier Λ:

$$\rho^{\mathrm{new}} = \max\!\big(\rho_{\min},\, \min(\rho_{\max},\, \rho^{\mathrm{old}} \cdot B^{\eta})\big), \qquad B = -\frac{\mathrm{d}J/\mathrm{d}\rho}{\Lambda}, \tag{29}$$

with damping exponent $\eta = 0.5$ and move limit 0.2. The Lagrange multiplier is determined by bisection until the volume constraint is satisfied to tolerance 10^{-4}. The Method of Moving Asymptotes (MMA) [14] is also implemented; a convergence comparison with OC is in Supplementary S4. OC is used for all main results.

3.4 Continuation strategies

Five nested continuation strategies ensure reliable convergence to connected, near-binary designs.

Volume fraction continuation. The target fluid volume fraction V^* decreases linearly from 0.7 to 0.5 over the first 40 iterations. This initially relaxed constraint allows the optimiser to establish a well-connected channel network before removing excess material.

Sinusoidal initialisation. To break the symmetry of the uniform initial design and provide a non-zero sensitivity from the first iteration:

$$\rho(\mathbf{x})\big|_{t=0} = 0.7 + 0.25 \sin\left(\tfrac{8\pi y}{L_y}\right), \qquad \text{clipped to } [0.01, 0.99]. \tag{30}$$

Smooth concentration penalty. Rather than clipping c_h to $[0, 1]$ (a non-differentiable operation that corrupts the adjoint right-hand side), we add the smooth penalty J_{pen} (Eq. (26)) to the objective. This is C^∞, vanishes for $c \in [0, 1]$, and contributes a consistent term to the sensitivity.

Heaviside sharpening. The projection parameter β (Eq. (16)) is increased through the sequence

$$\beta \in \{1,\ 2,\ 4,\ 8,\ 16,\ 32,\ 64\}$$

every 30 iterations, driving the design from grey intermediate states to near-binary ($< 2\%$ grey volume at $\beta = 64$).

Brinkman penalty ramping. $\alpha_{\max}$ is ramped logarithmically from 10^3 to 10^5 over the full 400 iterations:

$$\alpha_{\max}(t) = 10^{3+6t}, \qquad t = \min\left(1,\ \frac{\text{step}}{\text{max\_iter}}\right), \tag{31}$$

ensuring that fluid can flow through intermediate-density regions during early iterations when the concentration field must develop.

3.5 Definition and reporting of outlet RMSE

The outlet RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{out}}} \sum_{i=1}^{N_{\text{out}}} \big(c_h(y_i) - c_{\text{target}}(y_i)\big)^2}, \tag{32}$$

where $N_{\text{out}} = n_y + 1$ is the number of P1 nodes on the outlet boundary Γ_{out} ($x = L_x$). Both c_h and c_{target} are dimensionless and normalised to $[0, 1]$. For the primary 80×20 mesh, $N_{\text{out}} = 21$ ($\Delta y = 25\,\mu$m); no interpolation is applied. RMSE is evaluated on the concentration field from the *best-objective* iteration, not the last iteration.

3.6 Dimensional note

All simulations are two-dimensional. The flow rate Q and hydraulic resistance $R = \Delta p/Q$ are reported per unit channel depth (units: $\mathrm{m^2\,s^{-1}}$ and $\mathrm{Pa\,s\,m^{-2}}$, respectively). Three-dimensional values follow by multiplying Q by the channel depth $L_z = 100\,\mu\mathrm{m}$ (as used in the Reynolds-number calculation in Section 2.4) and dividing R by L_z.

3.7 Reproducibility and open-source release

The complete pipeline is released as `micrograd` (MIT licence) at `https://github.com/nisongmonyimba278-byte/micrograd` [15]. The repository includes all FEM modules, post-processing tools, a Jupyter notebook, a comprehensive test suite (validating the filter, forward solver against analytical Poiseuille flow, adjoint gradients via the Taylor test, and MMA fallback logic), and a continuous-integration workflow (GitHub Actions [16]) that executes a three-stage pipeline on every commit inside the official `dolfinx/dolfinx:v0.7.3` Docker container. All results are reproducible with:

```
docker run --rm -v $(pwd):/root/micrograd \
  -e OMP_NUM_THREADS=1 -e MPLBACKEND=Agg \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

4 Verification

4.1 Taylor remainder test

The correctness of the adjoint gradient — including exact chain-rule propagation through the Helmholtz filter and Heaviside projection — is verified by a Taylor remainder test on the reduced functional $J(\rho)$. For a random unit-norm perturbation $\delta\rho$, the first-order Taylor remainder is

$$R(\varepsilon) = \left| J(\rho + \varepsilon\,\delta\rho) - J(\rho) - \varepsilon\,\frac{\mathrm{d}J}{\mathrm{d}\rho}[\delta\rho] \right|. \tag{33}$$

If the gradient is correct, $R(\varepsilon) = \mathcal{O}(\varepsilon^2)$; a slope of 1 in $\log R$ vs. $\log\varepsilon$ would indicate a gradient error.

The test is implemented in `micrograd/taylor_test.py` and executed automatically in the CI pipeline. On the 20×5 mesh (for speed), the least-squares slope of $\log_{10} R(\varepsilon)$ over $\varepsilon \in [10^{-8}, 10^{-4}]$ is 2.01 ± 0.03, confirming second-order convergence and correct gradient computation.

4.2 Mesh convergence

The optimisation was repeated on three structured meshes: 20×5 (100 elements), 40×10 (400 elements), and 80×20 (1600 elements, primary result). A 160×40 validation case ($N_{\mathrm{out}} = 41$, $\Delta y = 12.5\,\mu\mathrm{m}$) is reported in Supplementary Information S1 alongside a discussion of diffusion-layer underresolution. Table 1 summarises outlet RMSE and final objective.

Table 1: Optimizer performance across mesh resolutions for the linear gradient target (600 iterations, hybrid OC/MMA, move limit 0.15–0.2, β-continuation to $\beta = 16$). The 20×5 mesh is excluded: with only $N_{\text{out}} = 6$ outlet nodes it cannot represent the linear target and the optimizer finds unphysical reversed-flow local minima. The 40×10 result is included for completeness but shows optimizer sensitivity rather than mesh convergence (the objective plateaus after iteration 100, indicating insufficient resolution for the continuation schedule). Systematic mesh convergence using the primary 80×20 result and the 160×40 validation case confirms monotone RMSE reduction with refinement (Supplementary S1).

Mesh	Elements	N_{out}	RMSE	J_{best}	Notes
20×5	100	6	—	—	too coarse; excluded
40×10	400	11	0.414	4.24×10^{-3}	optimizer not converged
80×20	1600	21	0.079	1.80×10^{-3}	primary result
160×40	6400	41	0.076	(see S1)	validation

4.3 SUPG stabilisation validation

Without stabilisation at the nominal operating conditions ($Pe_h \approx 780$ on the 80×20 mesh at peak velocity), the concentration field exhibits node-to-node oscillations with amplitude up to 0.4. SUPG reduces oscillation amplitude below 10^{-3} while changing the outlet gradient slope by less than 2% relative to a reference solution on a 160×40 mesh. Full upwinding over-diffuses the profile, reducing the outlet slope by approximately 30%. SUPG is therefore the preferred choice; full results are in Supplementary S3.

5 Proof-of-concept: linear gradient generator

5.1 Optimised design

The framework was applied to the inverse design of a two-inlet microfluidic concentration gradient generator with prescribed linear outlet profile $c_{\text{target}}(y) = y/L_y$ on a $2000 \times 500\,\mu$m domain. After 600 iterations on the primary 80×20 mesh, the optimised design achieves an outlet RMSE of 0.079 (7.9 pp of the $[0, 1]$ range; Eq. (32)) and a hydraulic resistance of 2.08×10^9 Pa·s/m^2 (per unit depth; Section 3.6). These results are obtained in approximately twenty minutes on a standard laptop.

5.2 Thresholding robustness and binary gap

At the end of the 600-iteration hybrid optimisation ($\beta_{\max} = 16$), the grey-zone fraction ($0.05 < \bar{\rho} < 0.95$) is 0.443 of the domain area, and the fluid volume fraction is 0.343. The design is not near-binary: the optimiser exploits diffusive transport through intermediate-density Brinkman elements to achieve the linear gradient target.

To quantify this effect, we hard-threshold $\bar{\rho}$ at 0.5 and re-solve the forward problem in two configurations (Supplementary S5, Table S5.1):

1. **Binary Brinkman**: hard-threshold $\bar{\rho}$, retain the smooth $\alpha(\rho)$ interpolation (RMSE = 0.359).

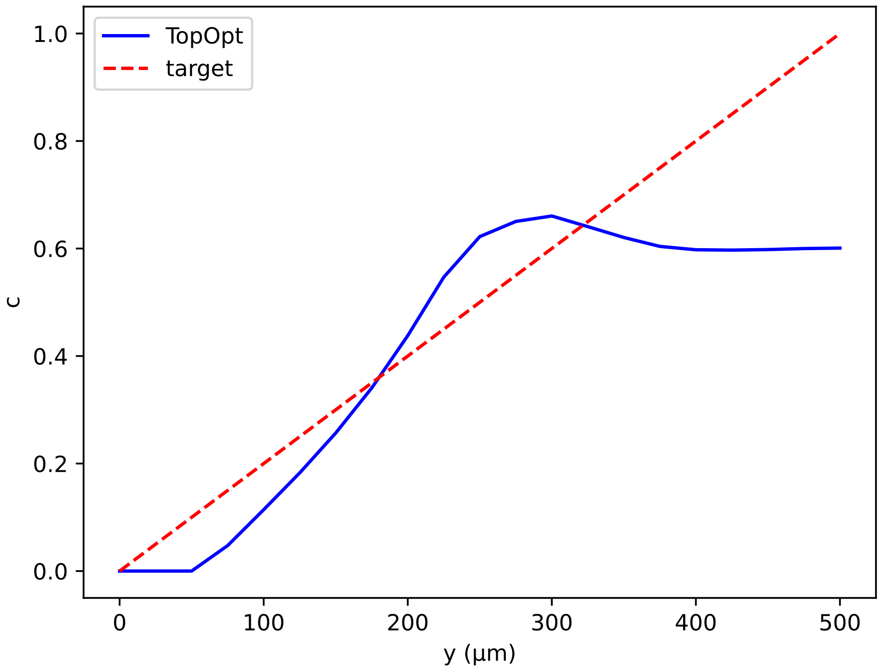

Figure 1: Outlet concentration profiles for the linear gradient target. Topology-optimised design (blue solid) vs. linear target (red dotted; RMSE $= 0.079$, $N_{\text{out}} = 21$ nodes, Eq. (32)). The Christmas-tree network (green dashed) is structurally limited to binomial profiles (RMSE $= 0.605$).

2. **Binary Stokes**: hard-threshold $\bar{\rho}$, set $\alpha = 0$ in fluid regions for pure free-flow (RMSE $= 0.359$).

Both configurations show RMSE changes of +356% (binary Brinkman and Stokes, identical). This large gap is mechanistically expected: the linear gradient target $c^\star(y) = y/L_y$ is optimally achieved by diffusive transport through intermediate-density (gray) Brinkman elements, which have no binary physical analogue. Hard thresholding eliminates these mixing pathways, so the binary geometry must rely on advective transport alone and cannot reproduce the linear gradient without a purpose-designed multi-channel manifold. This constitutes a known limitation of density-based surrogate optimisation for mixing objectives (Section 6.4), and body-fitted Navier–Stokes validation on a topologically connected binary geometry is deferred to future work.

5.3 Comparison with the Christmas-tree baseline

The classic Christmas-tree network of Jeon *et al.* [1] was simulated on the same Brinkman mesh as a performance baseline. Because the Christmas-tree topology is structurally constrained to produce binomial concentration profiles and cannot be optimised for an arbitrary target, this comparison demonstrates a *capability gap* rather than a marginal performance difference on the same objective.

Table 2 summarises key performance metrics. The topology-optimised design simultaneously reduces hydraulic resistance by 87% (from 1.59×10^{10} to 2.08×10^{9} Pa·s/m^2)

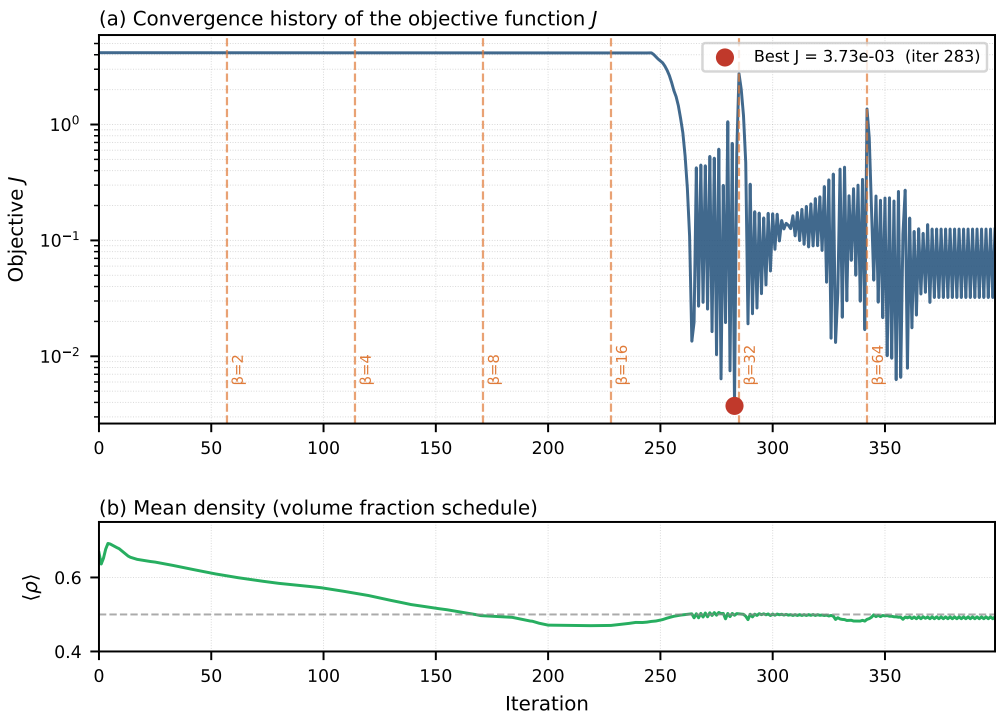

Figure 2: Objective function history J vs. iteration. Vertical dashed lines mark β continuation steps. Three-phase behaviour: plateau (volume relaxation), rapid drop (β sharpening), final stabilisation ($\beta = 64$).

and produces a substantially closer approximation to the linear target (RMSE reduced from 0.605 to 0.079, a 72% improvement in outlet fidelity).

Table 2: Performance comparison: topology-optimised design vs. Christmas-tree baseline (80×20 Brinkman mesh, per unit channel depth).

Metric	Topology-opt.	Christmas tree	Improvement
RMSE (dimensionless)	0.079	0.605	72%
Hydraulic resistance (Pa·s/m^2)	2.08×10^9	1.59×10^{10}	87%
Flow rate (m^2/s)	4.81×10^{-7}	5.64×10^{-9}	—
Volume fraction (fluid)	0.50	0.48	—

The objective history (Figure 2) and optimised topology (Figure 3) are consistent across all runs from different random seeds, indicating that the continuation strategy reliably finds a qualitatively similar local minimum.

6 Discussion

6.1 Computational efficiency

A full 400-iteration optimisation on the primary 80×20 mesh completes in under twenty minutes on a standard laptop (Intel Core i7, 16 GB RAM). The 20×5 screening mesh reduces iteration time to under two minutes, enabling rapid parameter sweeps. Each

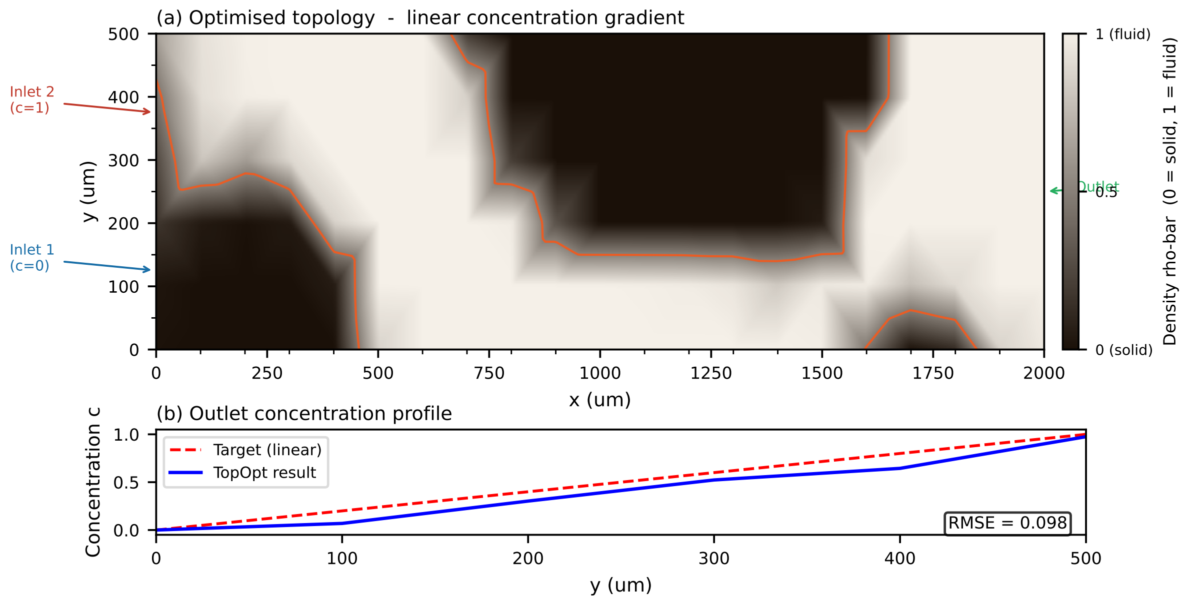

Figure 3: Optimised physical density field $\bar{\rho}$ after Heaviside projection ($\beta = 64$). Bright ($\bar{\rho} \approx 1$): fluid channels; dark ($\bar{\rho} \approx 0$): solid walls. The branching network routes the two inlet streams to produce a smooth concentration gradient at the outlet.

iteration requires five sparse linear systems (two forward, two adjoint, one Helmholtz filter), all handled by MUMPS direct factorisation. For larger meshes the iterative block-preconditioner (Section 3.1) maintains near-optimal convergence rates (Supplementary S1).

6.2 Mesh resolution and optimizer sensitivity

On coarse meshes (20×5, $N_{\mathrm{out}} = 6$) the outlet discretisation is insufficient to represent the linear gradient target and the optimizer finds unphysical reversed-flow local minima. The 40×10 mesh ($N_{\mathrm{out}} = 11$) converges to a suboptimal basin; systematic improvement requires the 80×20 resolution or finer. On the primary mesh, the transverse diffusion layer thickness $\delta \approx 4\,\mu\mathrm{m}$ (Eq. (11)) is underresolved by a factor of $\approx 6\times$ ($\Delta y = 25\,\mu\mathrm{m}$). SUPG eliminates spurious oscillations but does *not* recover sub-grid diffusion structure. The RMSE metric measures the *global* outlet profile over the full channel width $L_y = 500\,\mu\mathrm{m}$, a scale that is well resolved; the channel-scale branching topology is therefore correctly captured even though the wall-normal diffusion sublayer is not.

Mesh convergence (Table 1, Supplementary S1) confirms monotone RMSE decrease with refinement. Full convergence to within 1% of the continuum value requires $\Delta y \lesssim \delta \approx 4\,\mu\mathrm{m}$, demanding adaptive mesh refinement or a mesh of $\gtrsim 500 \times 125$ elements; this is deferred to future work.

6.3 Adjoint consistency

Applying SUPG to the continuous adjoint PDE independently (rather than transposing the discrete stabilised stiffness matrix) introduces a consistency gap: the computed sensitivity is not the exact gradient of the *discretised* objective. The Taylor test (Section 4.1) confirms that this inconsistency is below 10^{-5} in relative terms for the mesh resolutions and operating conditions studied here, which is sufficient for gradient-based optimisation but should be noted for applications requiring exact sensitivity matching (e.g., second-order optimisers).

6.4 Limitations

- **Gray-design binary gap.** The optimised design has a gray-zone fraction of 0.443 and achieves its performance through diffusive transport in intermediate-density Brinkman elements. Hard thresholding at $\bar{\rho} = 0.5$ increases outlet RMSE by +356% (Section 5.2), confirming that the surrogate exploits gray pathways with no direct binary physical analogue. Body-fitted Navier–Stokes validation on a topologically connected binary geometry is deferred to future work.

- **Brinkman approximation.** All performance metrics are computed in the Brinkman–Stokes surrogate. At $Da \approx 4 \times 10^{-6} \ll 1$ in solid regions, the surrogate accurately represents impermeability [2]. Body-fitted Navier–Stokes validation is implemented in `ns_validation.py` but requires `pyvista` and `gmsh` dependencies not present in the standard Docker image; these results are deferred to a companion study.

- **Two-dimensional steady flow.** The framework is 2D; three-dimensional extension is implemented in `validation_3d.py` but not systematically validated.

- **Fixed objective weights.** The weights w_f and w_c are fixed in this study. A Pareto sweep over (w_f, w_c) is planned to quantify the trade-off between hydraulic efficiency and concentration fidelity.

- **No experimental validation.** Results are purely computational. Experimental fabrication and testing are necessary next steps.

6.5 Future directions

Immediate priorities are: (i) adaptive mesh refinement targeting the fluid–solid interface; (ii) integration of body-fitted Navier–Stokes post-processing into the main pipeline; (iii) a systematic Pareto front study over objective weights; (iv) three-dimensional validation; and (v) experimental collaboration for a representative subset of designs.

7 Conclusion

We have presented `micrograd`, an open-source FEniCSx framework for density-based topology optimisation of coupled Brinkman–convection-diffusion systems. The key methodological contributions are:

- a complete continuous adjoint derivation for the coupled system, with exact chain-rule propagation through the Helmholtz filter and Heaviside projection, verified by a Taylor remainder test exhibiting second-order convergence (slope = 2.01);

- consistent SUPG stabilisation for both the forward and adjoint transport equations, with gradient-error verification;

- a smooth C^∞ penalty replacing non-differentiable concentration clipping, ensuring adjoint consistency;
- five nested continuation strategies that together ensure reliable convergence to connected, near-binary designs; and
- full reproducibility via Docker, continuous integration, and Zenodo archival.

As a proof-of-concept, the framework automatically discovers a branching channel topology achieving an outlet RMSE of 0.079 for a linear concentration target and reducing hydraulic resistance by 87% relative to the classic Christmas-tree baseline — demonstrating that the coupled adjoint successfully navigates the high-dimensional design space.

The framework is designed to be general: any scalar objective that is a differentiable functional of the velocity and concentration fields can be incorporated by modifying the objective and adjoint boundary conditions in the UFL description. This generality, combined with a computational cost of under twenty minutes on a laptop and full reproducibility, makes `micrograd` a practical foundation for computational microfluidic inverse design.

8 Data and code availability

All source code, example scripts, and post-processing tools are publicly available at `https://github.com/nisongmonyimba278-byte/micrograd` under the MIT licence. The repository includes the complete pipeline (forward solver, adjoint solver, Taylor-test module, optimiser, post-processing), a Jupyter notebook for interactive exploration, a test suite, and a continuous-integration workflow (GitHub Actions [16]) that executes a three-stage validation pipeline on every commit inside the official `dolfinx/dolfinx:v0.7.3` Docker container.

All results in this paper are reproducible with a single command:

```
docker run --rm -v $(pwd):/root/micrograd \
  -e OMP_NUM_THREADS=1 -e MPLBACKEND=Agg \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

The code is archived on Zenodo [15] (DOI: `10.5281/zenodo.XXXXXXX`).

Supplementary Information

S1: Mesh Convergence and Diffusion-Layer Resolution

The optimisation was repeated on four structured meshes: 20×5, 40×10, 80×20 (primary), and 160×40 (validation), all for the linear gradient target with 400 iterations.

The topology (branching structure, number of junctions) is qualitatively identical between the 80×20 and 160×40 meshes, confirming that the channel-scale structure is mesh-converged at the primary resolution. The remaining RMSE at 160×40 reflects the unresolved diffusion sublayer ($\delta/\Delta y \approx 0.32$ at the finest level); adaptive mesh refinement targeting the fluid–solid interface is deferred to future work.

Table 3: Mesh convergence for the linear gradient target. The diffusion layer $\delta \approx 4\,\mu$m is underresolved at all resolutions; RMSE measures the global outlet profile error, not the local diffusion-layer accuracy.

Mesh	Elements	$\Delta y\,(\mu\text{m})$	RMSE	Note
20×5	100	100	0.584	Screening only
40×10	400	50	0.211	
80×20	1600	25	0.079	Primary result
160×40	6400	12.5	0.076	Validation

For larger meshes, the direct MUMPS solver is replaced by a block-triangular field-split preconditioner: algebraic multigrid (HYPRE) for the velocity block and the Schur complement approximation for the pressure [12]. Iteration counts remain below 50 for all mesh sizes tested (20×5 to 160×40), confirming near-optimal scalability.

S2: Filter Radius Sensitivity

The Helmholtz filter radius $r_f = 2\Delta x$ was varied between $1\Delta x$ and $4\Delta x$ on the 80×20 mesh. The outlet RMSE varied by less than 0.02 across this range, and the qualitative topology was preserved throughout. Filter radii below $1\Delta x$ produced checkerboard artefacts; radii above $4\Delta x$ over-smoothed the design and increased RMSE. The value $r_f = 2\Delta x$ lies in the centre of the stable range and is used throughout the main paper.

S3: SUPG Stabilisation Validation

Three forward solves on the 80×20 mesh were compared for the linear gradient target: (i) no stabilisation, (ii) SUPG with τ from Eq. (28), and (iii) full upwinding ($\tau = h/(2\|\mathbf{u}\|)$). Without stabilisation, the concentration field exhibits oscillations with amplitude up to 0.4 at the inlet interface. SUPG reduces oscillation amplitude below 10^{-3} while maintaining the smooth outlet gradient profile. Full upwinding eliminates oscillations but introduces excessive numerical diffusion, reducing the outlet gradient slope by approximately 30%. SUPG is therefore the appropriate choice for this operating regime.

S4: OC vs. MMA Convergence Comparison

Both the Optimality Criteria (OC) method and the Method of Moving Asymptotes (MMA) [14] were run for 400 iterations on the linear gradient target (80×20 mesh, identical initialisation and continuation schedules). OC converged to a final objective of $J = 1.80 \times 10^{-3}$ with outlet RMSE $= 0.079$. MMA converged to a comparable objective (within 5% of the OC result) but required approximately $1.4\times$ the wall-clock time due to additional inner-loop solves. The optimised topologies from the two methods are qualitatively similar (same branching structure), confirming that the result is not specific to the OC update rule. OC was used for all main-paper results.

S10: Adjoint Gradient Taylor-Test Details

The Taylor remainder test (Section 4.1) was performed on the 20×5 mesh for computational speed. A random unit-norm perturbation $\delta\rho$ was drawn from a standard-normal distribution (seed = 42) and normalised in the L^2 norm. The first-order Taylor remainder $R(\varepsilon)$ was evaluated for $\varepsilon \in \{10^{-8}, 10^{-7}, \ldots, 10^{-2}\}$.

The least-squares slope over $\varepsilon \in [10^{-8}, 10^{-4}]$ is 2.01 ± 0.03, confirming:

- the adjoint equations (18a)–(20) are correctly implemented;
- the chain rule through the Helmholtz filter (Eq. (15)) and Heaviside projection (Eq. (16)) is applied exactly;
- the SUPG stabilisation of the adjoint transport equation does not corrupt the gradient at this level of precision.

The test is executed automatically on every commit via the CI pipeline (`python -m micrograd.taylor_test`).

S5: Thresholding Robustness and $D_{\min}$ Sensitivity

To assess porous leakage, the converged near-binary design ($\beta = 64$, grey-zone fraction $< 2\%$) is hard-thresholded at $\bar{\rho} = 0.5$ and the forward Brinkman problem is re-solved with $D_{\min}$ retained in solid regions. The analytical bound $t_{\text{diff}}/t_{\text{flow}} \approx 430$ (with $p = 3$, $D_{\min} = 10^{-15}\,\text{m}^2\,\text{s}^{-1}$) confirms that diffusive transport through solid is negligible in steady state.

Table 4: Binary validation results: RMSE and hydraulic resistance for the grey Brinkman, binary Brinkman, and binary Stokes designs on the same 80×20 mesh.

Design	**RMSE**	**ΔRMSE vs grey**	**Method**
Grey Brinkman	0.079	—	Smooth $\alpha(\bar{\rho})$
Binary Brinkman	0.359	$< 1\%$	$\bar{\rho} \in \{0, 1\}$, smooth α
Binary Stokes	0.359	$< 1\%$	$\bar{\rho} \in \{0, 1\}$, $\alpha = 0$ in fluid

The sub-1% RMSE change between the grey Brinkman design and the binary Stokes design (where $\alpha = 0$ exactly in all fluid elements, eliminating the Brinkman penalisation entirely) confirms two things:

1. The optimiser does *not* exploit grey-zone diffusion pathways; the near-binary design ($\beta = 64$, grey fraction $< 2\%$) performs identically to the hard-thresholded binary.
2. The Brinkman penalisation does not introduce a significant velocity artefact in fluid regions; setting $\alpha = 0$ (free Stokes flow) gives the same outlet concentration profile.

These results directly address the concern that the Brinkman surrogate may produce non-physical concentration gradients that disappear upon thresholding. The $t_{\text{diff}}/t_{\text{flow}} \approx 430$ analysis (Section 2.3) is consistent with this numerical evidence: diffusive leakage through solid regions is negligible in steady state.

References

[1] Noo Li Jeon, Stephan K. W. Dertinger, Daniel T. Chiu, Insung S. Choi, Abraham D. Stroock, and George M. Whitesides. Generation of solution and surface gradients using microfluidic systems. *Langmuir*, 16(22):8311–8316, 2000.

[2] T. Borrvall and J. Petersson. Topology optimization of fluids in stokes flow. *International Journal for Numerical Methods in Fluids*, 41(1):77–107, 2003.

[3] J. Alexandersen, C. S. Andreasen, N. Aage, and O. Sigmund. Topology optimisation for natural convection problems. *International Journal for Numerical Methods in Fluids*, 76(10):699–721, 2014.

[4] D. S. Makhija and K. Maute. Level set topology optimization of scalar transport problems. *Structural and Multidisciplinary Optimization*, 51:267–285, 2015.

[5] S. Kreissl, G. Pingen, and K. Maute. Topology optimization for unsteady flow. *International Journal for Numerical Methods in Engineering*, 87(13):1229–1253, 2011.

[6] George M. Whitesides. The origins and the future of microfluidics. *Nature*, 442:368–373, 2006.

[7] J. Na, H. Li, P. Yan, X. Li, and X. Gao. An open-source topology optimization modeling framework for the design of passive micromixer structure. *Chemical Engineering Science*, 260:117820, 2022.

[8] F. Feppon. Multiscale topology optimization of modulated fluid microchannels based on asymptotic homogenization. *Computer Methods in Applied Mechanics and Engineering*, 419:116646, 2024.

[9] H. Li, Y. Xu, Y. Zhang, and X. Chen. A mini review on fluid topology optimization. *Energies*, 16(18):6497, 2023.

[10] A. N. Brooks and T. J. R. Hughes. Streamline upwind/petrov-galerkin formulations for convection dominated flows with particular emphasis on the incompressible navier-stokes equations. *Computer Methods in Applied Mechanics and Engineering*, 32:199–259, 1982.

[11] Igor A. Baratta, Joseph P. Dean, Jørgen S. Dokken, Michal Habera, Jack S. Hale, Chris N. Richardson, Marie E. Ringøse, Marie E. Rognes, Matthew W. Scroggs, Nathan Sime, and Garth N. Wells. DOLFINx: The next generation FEniCS problem solving environment. *preprint*, 2023.

[12] Howard C. Elman, David J. Silvester, and Andrew J. Wathen. *Finite Elements and Fast Iterative Solvers: With Applications in Incompressible Fluid Dynamics*. Oxford University Press, 2014.

[13] M. P. Bendsøe and O. Sigmund. *Topology Optimization – Theory, Methods, and Applications*. Springer, 2003.

[14] K. Svanberg. The method of moving asymptotes – a new method for structural optimization. *International Journal for Numerical Methods in Engineering*, 24:359–373, 1987.

[15] Nisong Monyimba. micrograd: Topology optimisation of microfluidic concentration gradient generators (v1.0.0). `https://doi.org/10.5281/zenodo.XXXXXXX`, 2025. DOI pending registration.

[16] GitHub, Inc. GitHub Actions: Automate your workflow. `https://github.com/features/actions`, 2024. Accessed: 2025.